

Relativistic Time Dilation & Coordinate Phase Integration

Complete Spacetime Mathematics of the Time Dilation DAW Engine

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August 3, 2026

Abstract

This document details the mathematical framework governing relativistic coordinate time dilation, Lorentz velocity transformations, Schwarzschild gravitational redshift event horizons, metric time grid quantization, universal node `timeIn` propagation, and digital audio phase integration in **Time Dilation DAW (v0.0.1)**.

1 Introduction to Relativistic Coordinate Time (τ)

In standard Digital Audio Workstations (DAWs), digital time progresses linearly at a constant sample rate f_s :

$$t(n) = \frac{n}{f_s} \quad (1)$$

In **Time Dilation DAW**, coordinate time is modulated by a dynamic relativistic factor $\gamma(t) \in [-16.0, 16.0]$, representing localized gravitational or velocity time dilation:

$$d\tau = \gamma(t) dt \quad (2)$$

The accumulated coordinate time $\tau(t)$ after physical elapsed timeline t is obtained by phase integration:

$$\tau(t) = \int_0^t \gamma(t') dt' \quad (3)$$

2 Universal Node `timeIn` Propagation Architecture

Every functional node object in the Time Dilation DAW graph engine incorporates an explicit or implicit Inlet 0 designated `timeIn` (`NodePortType::Time`). This architectural pattern guarantees universal relativistic time handling across all signal, control, telemetry, and sequencer nodes.

2.1 Hierarchical Time Dilation Inheritance

During each audio process cycle, the graph executes `propagateTimeDilationHierarchy()` prior to DSP evaluation:

$$\gamma_{\text{node}} = \begin{cases} \gamma_{\text{patched}} & \text{if Inlet 0 (`timeIn`) is explicitly connected to a time output} \\ \gamma_{\text{upstream}} & \text{if `timeIn` is disconnected (inherits upstream coordinate time)} \\ 1.0 & \text{if unpatched and root level (stationary reference frame)} \end{cases} \quad (4)$$

2.2 Unified Sample-Rate & Phase Modulation

Regardless of node class (Audio, Control, Monitor, or Sequencer), the effective local time step $d\tau_n$ evaluated by the node at sample step n is:

$$d\tau_n = \frac{\gamma_{\text{node}}[n]}{f_s} \quad (5)$$

For an audio node, $d\tau_n$ controls the Hermite sub-sample interpolation delay pointer advance rate. For a control or clock node (e.g. `seq`, `adsr~`, `phasor~`), $d\tau_n$ modulates the phase accumulator step $\Delta\phi = 2\pi f \cdot d\tau_n$.

3 Velocity Time Dilation (Special Relativity)

Under Einstein's Special Theory of Relativity, coordinate time dilation for an observer moving at relative velocity $v(t)$ with respect to speed of light c is defined by the Lorentz factor:

$$\gamma(v) = \frac{1}{\sqrt{1 - \frac{v(t)^2}{c^2}}} \quad (6)$$

For $v \rightarrow c$, $\gamma(v) \rightarrow \infty$, causing asymptotic time dilation. In our engine, γ is clamped to $[-16.0, 16.0]$ for real-time digital stability.

4 Gravitational Time Dilation & Event Horizon Redshift

Near a massive gravitational body of Schwarzschild radius $r_s = \frac{2GM}{c^2}$, time dilation at radius r follows:

$$\gamma(r) = \sqrt{1 - \frac{r_s}{r}} = \text{redshift} \quad (7)$$

As $r \rightarrow r_s^+$ (approaching the Event Horizon), $\text{redshift} \rightarrow 0$, causing gravitational time stasis ($\gamma \rightarrow 0.0$). In the `[time.singularity~]` node, non-linear curvature warping is applied:

$$\gamma_{\text{singularity}}(r, M) = \text{clamp} \left(\sqrt{\left| 1 - \frac{2GM}{c^2 r} \right|}, 0.0, 16.0 \right) \quad (8)$$

5 Retrograde Time Reversal & Metric Grid Quantization

5.1 Retrograde Time Reversal (`[time.retro~]`)

When $\gamma(n) < 0$, the coordinate time rate reverses ($\dot{\tau} = -1.0$), executing authentic *Retrograde Time Reversal*:

$$\tau[n] = \tau[n-1] + \frac{\gamma[n]}{f_s}, \quad \text{where } \gamma[n] \in [-16.0, 0.0] \quad (9)$$

Under retrograde flow, phase increments $\Delta\phi(n) = \frac{2\pi f \gamma(n)}{f_s} < 0$, causing audio buffer read pointers to traverse backwards in real time.

5.2 Metric Time Grid Quantization (`[time.quantize~]`)

To enforce rhythmic micro-step stuttering, coordinate time $\tau(t)$ is discretized against a metric grid division $Q \in \mathbb{R}^+$ (e.g. 1/16-note grid duration $T_Q = \frac{60}{4 \cdot \text{BPM}}$):

$$\tau_{\text{quantized}}(t) = \lfloor \frac{\tau(t)}{T_Q} \rfloor \cdot T_Q + \text{hold}(\tau(t) \pmod{T_Q}) \quad (10)$$

6 Coordinate Phase Integration in Discrete Sampler Domain

In discrete time at sample rate f_s , the per-sample phase increment $\Delta\phi(n)$ for an oscillator or table reader at frequency f is:

$$\Delta\phi(n) = \frac{2\pi \cdot f \cdot |\gamma(n)|}{f_s} \quad (11)$$

The continuous phase accumulator $\phi[n]$ is updated according to:

$$\phi[n] = (\phi[n-1] + \Delta\phi(n)) \pmod{2\pi} \quad (12)$$

This guarantees phase continuity and exact frequency scaling under arbitrary dynamic time warping $\gamma(t)$.